

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall	Wall upgrade: • Exterior wall insulation: Expanded Polystyrene (EPS) Foam Board, target R-30.0
Scenario 2	roof	Roof upgrade:

		• Roof insulation: Blown Fiberglass, target R-30.0
Scenario 3	wall + roof + window + door	Wall upgrade: • Exterior wall insulation: Extruded Polystyrene (XPS) Foam Board, target R-10.0 (skipped: target R-value already satisfied) Roof upgrade: • Roof insulation: Blown Mineral Wool, target R-15.0 Window upgrade: • 1-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) Door upgrade: • Door replacement: glass door (2 of 3 door(s) skipped: door type incompatibility)

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	236.54	233.46	3.08	1.30%
Scenario 2	Total Site Energy (GJ)	236.54	234.42	2.12	0.90%
Scenario 3	Total Site Energy (GJ)	236.54	228.19	8.35	3.53%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison

Baseline \$9,841

Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison

Baseline 20,894

Scenario 3\$9,554

Baseline

Scenario 3

Max saving: **\$-287 (-2.92%)**

Scenario 320,320

Baseline

Scenario 3

Max saving: **-574 kg CO₂e (-2.75%)**

Min Retrofit Construction Cost

\$7,922

Scenario 2

Min Retrofit Embodied Carbon

1,577 kg CO₂e

Embodied carbon intensity (ECI): 3.08 kgCO₂e/m²
Scenario 1

Lowest Retrofit Cost Payback Period

75 years

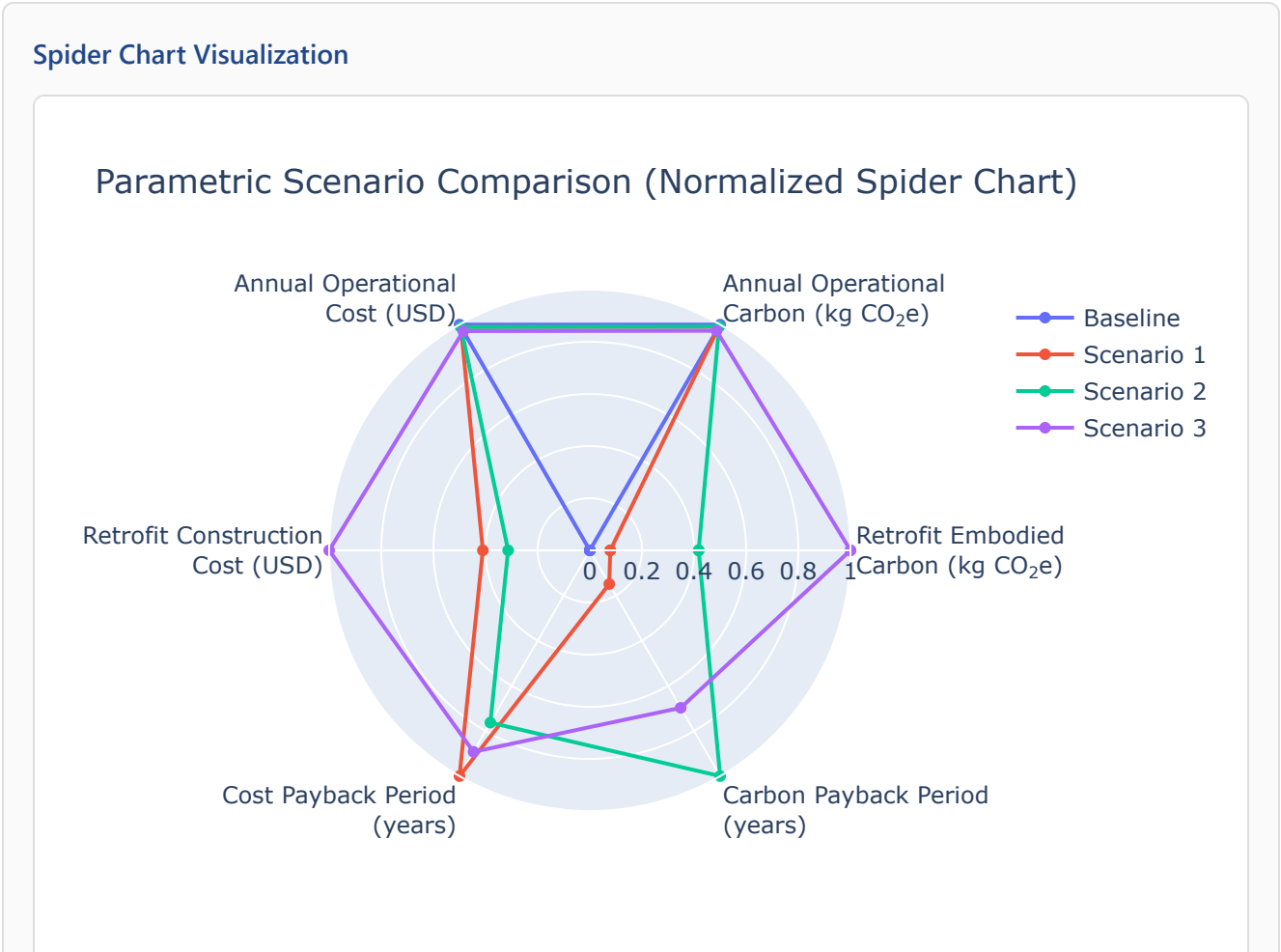
Scenario 2

Lowest Retrofit Carbon Payback Period

7.49 years

Scenario 1

Comparative Visualizations between Renovation Scenarios



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1		99 yrs
Scenario 2		75 yrs
Scenario 3		88 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1		7.49 yrs
Scenario 2		50 yrs
Scenario 3		35 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)	Product Lifetime (years)
Scenario 1	Wall insulation	Expanded Polystyrene (EPS) Foam Board	21	282	N/A	N/A	0.03	20	60
Scenario 2	Roof insulation	Blown Fiberglass	125	599	0.21	N/A	0.03	30	68
Scenario 3	Wall insulation	Extruded Polystyrene (XPS) Foam Board	N/A	N/A	N/A	N/A	0.03	35	75
Scenario 3	Roof insulation	Blown Mineral Wool	61	599	0.10	N/A	0.03	90	30
Scenario 3	Window frame	N/A	N/A	56	N/A	N/A	N/A	N/A	15
Scenario 3	Window caulking	N/A	0.0034	N/A	N/A	N/A	N/A	N/A	10

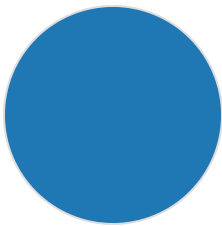
Scenario 3	Door	glass door	N/A	3.91	N/A	N/A	0.10	2,500	30
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Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

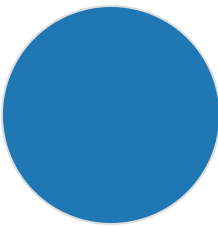
Cost breakdown by retrofit measure



■ Wall: \$10,372 (100.0%)

Total: \$10,372

Carbon breakdown by retrofit measure

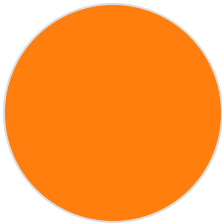


■ Wall: 1,577 (100.0%)

Total: 1,577 kg CO₂e

Scenario 2

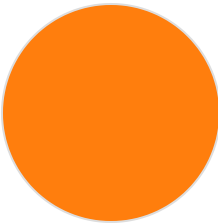
Cost breakdown by retrofit measure



■ Roof: \$7,922 (100.0%)

Total: \$7,922

Carbon breakdown by retrofit measure



■ Roof: 8,391 (100.0%)

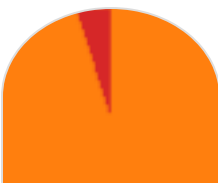
Total: 8,391 kg CO₂e

Scenario 3

Cost breakdown by retrofit measure



Carbon breakdown by retrofit measure



Roof: \$23,537 (93.2%)
Door: \$1,731 (6.8%)
Total: \$25,268

Roof: 19,122 (95.2%)
Door: 972 (4.8%)
Total: 20,095 kg CO₂e

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	20,894	\$9,841
Scenario 1	1,577	\$10,372	20,684	\$9,736
Scenario 2	8,391	\$7,922	20,727	\$9,736
Scenario 3	20,095	\$25,268	20,320	\$9,554